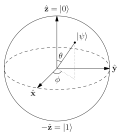
Quantum Computing

Instructions

- This **2 h** examination consists of **4** independent questions of equal ponderation.
- The use of a non-programmable calculator is permitted, but not especially helpful.
- You may write your answers in French or English (but not both).
- The purpose of this evaluation (just like any other) is to convince us that you are in capacity of saying sensible things on the topics at hand using the appropriate concepts ; please express yourself accordingly !
- Notation : * stands for complex conjugation, † for matrix conjugation and transposition.
- Some useful matrices : $X = \mathrm{NOT} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$, $S = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$, $H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$.



Question 1

a) The state of a *quantum trit* (or *qutrit*) can be described as a nonzero element in the complex vector space $\mathcal{V}_3$ with basis $|0\rangle$, $|1\rangle$ and $|2\rangle$. Suppose a qutrit is in state

$$|\phi\rangle = |0\rangle - 4i\,|1\rangle + 2\sqrt{2}\,|2\rangle$$

and a measurement is made in the computational basis. Describe what are the possible outcomes and their respective probabilities.

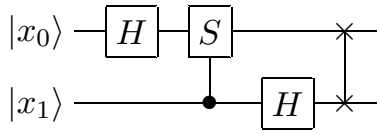
b) Let A be the 3×3 complex matrix

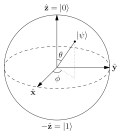
$$A = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1 & 1 \\ 1 & \omega & \omega^* \\ 1 & \omega^* & \omega \end{bmatrix} \qquad \text{where } \omega = e^{\frac{2\pi i}{3}} = -\frac{1}{2} + \frac{\sqrt{3}}{2}\,i.$$

[NB : here $\omega^* = \omega^{-1} = \omega^2$ since $\omega^3 = 1$]

Explain why we know that A can *a priori* be physically realized as a quantum gate that acts on qutrits.

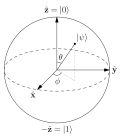
c) Write down the 4×4 unitary matrix corresponding to the following quantum circuit acting on two qubits by working out its output on each of the four basic states $|00\rangle$, $|01\rangle$, $|10\rangle$ and $|11\rangle$.





Question 2

Describe 3 of the desired properties for physical realizations of qubits (Di Vincenzo criteria) and their impact on the development of large-scale fault-tolerant quantum computers.

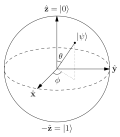


Question 3

a) Explain what a quantum algorithm is in the quantum circuit model and the way in which its complexity is measured.

b) Give an example of a problem for which quantum computers exhibit a quantum advantage over their classical counterpart (be specific).

c) "Quantum computers are just big computers that perform fast, Roger [1] says. Soon, as technology improves, everyone will be able to enjoy the benefits of quantum computing at home to perform their everyday tasks like word processing and watching videos." Comment !



Question 4

Context. You are responsible for designing a chip with 6 electronic components. The chip has two layers, and each component can be placed on one or the other. Some components lose their efficiency if they are placed on the same layer. The following graph shows the pairs of components for which the problem occurs.

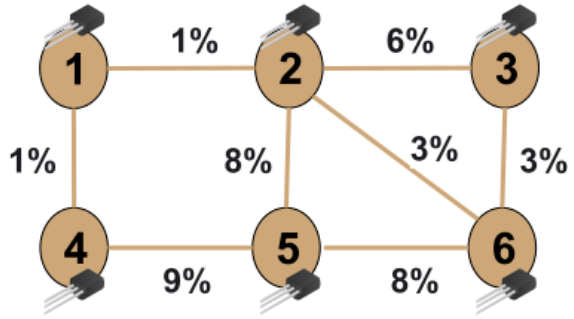


FIGURE 1 – Loss percentages between some pairs of components

3 examples :

- if the components (1) and (2) are on the same layer, the percentage of loss is 1,
- if (4) and (5) are on the same layer, the percentage of loss is 9,
- if (1) and (5) are on the layer, there is no loss.

Questions.

a) Trying to minimize the loss of efficiency of the chip, which optimization problem matches with your objective ?

b) Formalize the objective function of the problem based on the graph and using $\{-1, 1\}$ variables.

c) Try to find an optimal solution (by hand).

1. Any resemblance to an actual person, living or dead, is purely coincidental